University of
Inland Norway

ØKA2015 Applied Data Science

Autumn 2026 | 7.5 study points | Lillehammer

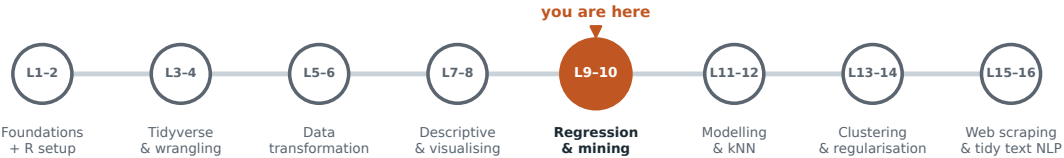
Lecture 10

Data mining and simple linear regression

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Thursday 22 October 2026 | 12:15–14:00

Where we are in the course



So far (Lectures 7 to 9)

You described data with numbers, including the correlation r, and showed it with `ggplot2`, including a trend line drawn by `geom_smooth()`.

Today

The simple linear regression model: a straight line that predicts one variable from another. How it is estimated, how it is interpreted, and what it cannot show.

Materials for this lecture

The work is done in **RStudio on your own laptop**, typing each command yourself. Everything below is also online.

Slides | this document, as a PDF

R script | `lecture10_regression.R`: every command from today, in today's order, each line explained in a comment. We run it in parallel.

R Markdown tutorial | `lecture10_regression.Rmd` and its knitted PDF: the same content as a document

Data | `trees`, which comes with R

Reading

Camm et al., *Business Analytics*, Ch. 8, sections 8.1 to 8.3 and 8.5 (Ch. 7 in the teaching plan, an older edition). Wright et al., *Tidyverse Skills*, Ch. 5.7.

Where it is

GitHub, the course repository, folder `lecture10`:
`github.com/touseefhameed/oka2015-applied-data-science`

Google Colab: the script in your browser, nothing to install. Open in Colab, also a badge on the GitHub page.

GitHub

Colab, Lecture 10

Today's plan

1. **Data mining**: describing and predicting
2. **Simple linear regression**: the model and the least squares line
3. **Fitting it in R**: `lm()`, the estimates, predictions
4. **How sure are we**: the standard error and the p-value
5. **How good is the line**: R-squared and the check plots
6. **Correlation and causation**

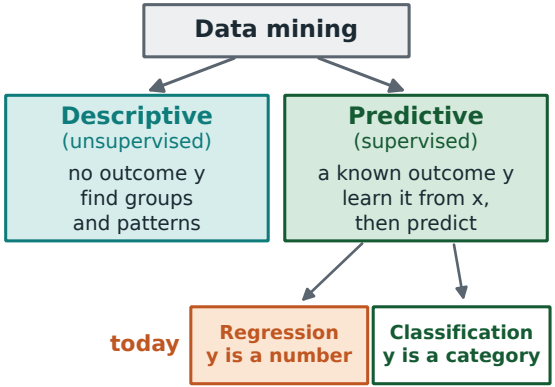
How the session runs

Each code slide corresponds to one block of the script. Run the block and compare the output with the slide.

By the end you will be able to

- ✓ tell predictive from descriptive data mining
- ✓ fit a straight line with `lm()` and draw it
- ✓ read the slope and the intercept in the units of the data
- ✓ read the standard error and the p-value, and decide
- ✓ judge the fit with R-squared and the check plots
- ✓ explain why association is not causation

1. Data mining: describing and predicting



Data mining | using software to find useful patterns in large amounts of data.

Descriptive | there is no answer y to predict; find groups and patterns. *Example:* a shop groups its customers by what they buy.

Predictive | learn from past cases where the answer y is known, then predict y for new cases. *Example:* the price of a house from its size.

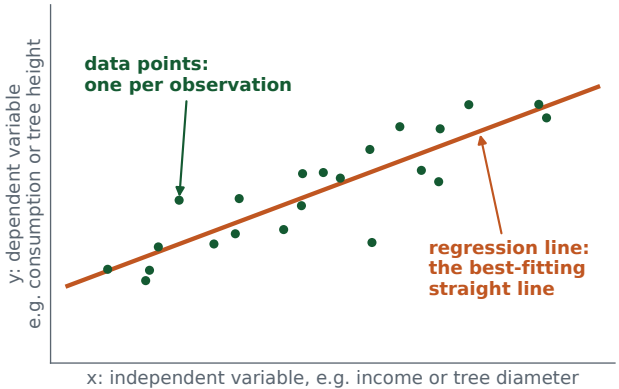
Regression or classification | y is a number (a price, a height) or a category (repays a loan or not).

The names

y: dependent variable, outcome, target. x: independent variable, predictor, feature.

Camm et al., *Business Analytics*, Ch. 1.3 (p. 4), Ch. 6 (p. 258), Ch. 8 (p. 394), Ch. 10 (p. 524)

2. What linear regression does



Camm et al., Ch. 8 (p. 394)

Linear regression | a straight line that predicts one variable from another.

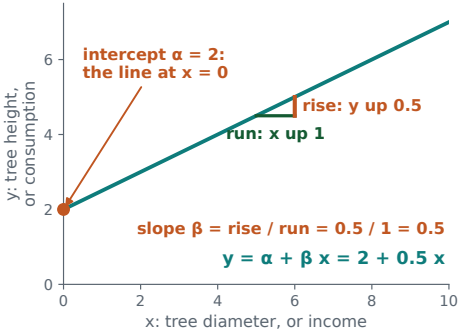
Dependent variable y | the variable we predict: consumption, or the height of a tree. On the vertical axis.

Independent variable x | the variable we predict it from: income, or the diameter of the tree. On the horizontal axis.

Simple linear regression | one x and a straight line. With several x it is *multiple* regression (next lecture).

2. The model

$$y = \alpha + \beta x + \varepsilon$$



α, the intercept | the value of y when $x = 0$, where the line crosses the y-axis. In the plot, 2.

β, the slope | rise over run, the change in y divided by the change in x. When x goes up by 1, y goes up by β on average. In the plot, 0.5. *Example:* consumption $= \alpha + \beta \times$ income $+\varepsilon$: β is how much consumption rises when income rises by one unit.

ε, the error term | everything else that affects y, which the line does not capture. For a tree: age, soil, light.

α and β | the true values for the whole population, which we never know. From a sample we estimate them: $\hat{\alpha}$ and $\hat{\beta}$ ("alpha-hat", "beta-hat").

Camm et al., Ch. 8.1 (the book writes β_0 for α and β_1 for β). *Simple*: one x. *Linear*: a straight line.

2. Today's data: 31 black cherry trees

```
data(trees)
glimpse(trees)
```

```
Rows: 31
Columns: 3
$ Girth  <dbl> 8.3, 8.6, 8.8, 10.5, ...
$ Height <dbl> 70, 65, 63, 72, 81, ...
$ Volume <dbl> 10.3, 10.3, 10.2, 16.4, ...
```

`trees` comes with R: nothing to install or download.

Girth | the trunk *diameter* in inches, 4.5 feet above the ground. The name is misleading; the help page `?trees` says so.

Height | in feet. 1 inch = 2.54 cm, 1 foot = 0.30 m.

Volume | timber in cubic feet.

The question

The diameter of a tree is easy to measure; the height of a standing tree is not. **Can we predict height from diameter?**

$y =$ `Height`, $x =$ `Girth`

Wright et al., *Tidyverse Skills*, Ch. 5.7.4, asks the same question of the same data.

2. A first look: the scatter plot

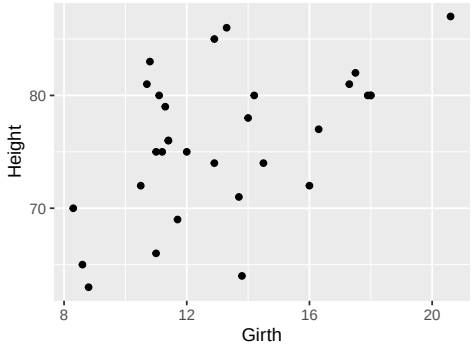
```
ggplot(trees, aes(x = Girth, y = Height)) +
  geom_point()
cor(trees$Girth, trees$Height)
```

```
[1] 0.5192801
```

Axes | x (diameter) across, y (height) up.

Pattern | the points go up: thicker trees tend to be taller, but with much scatter.

$r = 0.52$ | a positive, moderate correlation (Lecture 7).



Correlation measures how closely the points follow a straight line. Regression estimates the line itself and uses it to predict.

2. Direction and strength of a relationship



Direction | positive: y goes up with x. Negative: y goes down.

Strength | how close the points lie to a straight line. The correlation r measures it, from -1 to $+1$.

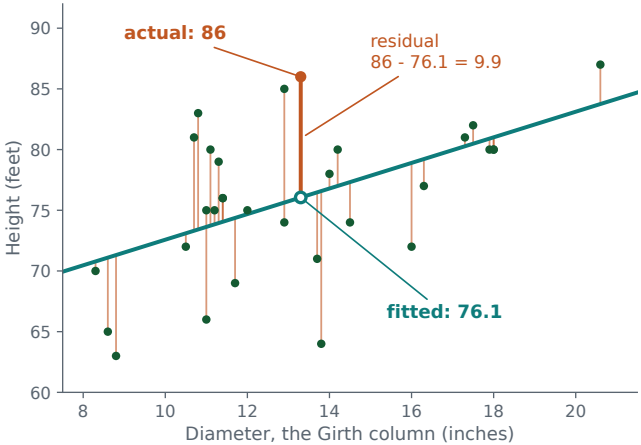
Our trees | $r = 0.52$: positive and moderate.

Key idea

A clear curve can have r near 0: r only measures straight-line patterns. Always plot the data first.

Camm et al., Ch. 2.8 *Measures of association between two variables*

2. Which line? The least squares method



Camm et al., Ch. 8.2 *Least squares method*

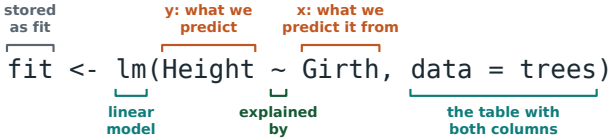
Fitted value $\hat{y}$ | the point on the line at a tree's diameter: the height the line says that tree should have.

Residual | actual minus fitted, $y - \hat{y}$: how far a point lies above or below the line. The tree marked is 9.9 feet above it.

Least squares | the line with the smallest sum of squared residuals. The residuals are squared so that the ones above the line (positive) and below it (negative) do not cancel.

In R | `lm()` finds this line.

3. Fit the model with `lm()`




```
fit <- lm(Height ~ Girth, data = trees)
fit
```

```
Coefficients:
(Intercept)        Girth
     62.031        1.054
```

Wright et al., Ch. 5.7.5 *Fitting the model*

General form | `lm(y ~ x, data = table)`. `lm()` finds the least squares line.

Output | `(Intercept)` is $\hat{\alpha}$, `Girth` is $\hat{\beta}$.

3. The estimates replace α and β

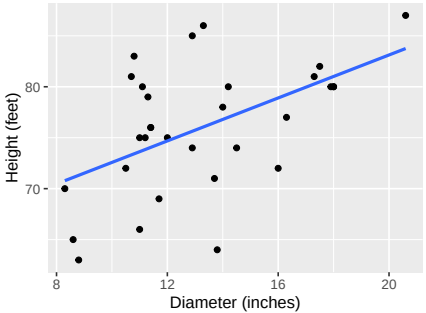
$$y = \alpha + \beta x + \varepsilon \quad \text{becomes}$$

$$\text{Height} = \mathbf{62.03} + \mathbf{1.054} \times \text{Girth}$$

Slope $\hat{\beta} = 1.054$ | **one inch more diameter goes with 1.05 feet more height, on average.** The unit of the slope is feet per inch.

Intercept $\hat{\alpha} = 62.03$ | the height the line gives for a diameter of 0. No tree is that thin (the diameters run from 8.3 to 20.6 inches), so it has no real meaning; it only positions the line.

The error term | is not one number. What is left of it are the residuals: how far each tree is from the line.



The fitted line on the scatter plot

Camm et al., Ch. 8.2: the intercept often lies outside the data and then has no meaning.

3. Draw the line

`ggplot(trees, aes(...))` | the data, with diameter on x and height on y.

`geom_point()` | one dot per tree.

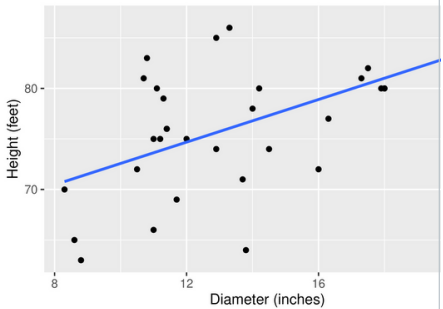
`geom_smooth(method = "lm")` | the straight line that `lm()` fits.

`se = FALSE` | the line only; the grey band comes in section 4.

`labs()` | the axis titles.

In RStudio | the message `geom_smooth() using formula = 'y ~ x'` is information, not an error.

```
ggplot(trees, aes(x = Girth, y = Height)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "Diameter (inches)", y = "Height (feet)")
```



The code and its plot in the knitted R Markdown tutorial

3. Fitted values, residuals and a prediction

```
trees_fit <- trees |>
  mutate(fitted   = fitted(fit),
         residual = residuals(fit))
head(trees_fit, 3)
```

```
  Girth Height Volume   fitted  residual
1   8.3     70   10.3 70.78257 -0.782575
2   8.6     65   10.3 71.09889 -6.098886
3   8.8     63   10.2 71.30976 -8.309759
```

```
predict(fit, newdata = tibble(Girth = 15))
```

```
       1
77.84685
```

Fitted value | the point on the line at a tree's diameter: the height the line says that tree should have. Tree 2: 71.1 feet.

Residual | how far the real height is from it. Tree 2 is 65 feet tall, so its residual is −6.1: 6.1 feet below the line.

`predict()` | the line's answer for a new tree: a tree 15 inches across is predicted to be 77.8 feet tall.

Only inside the data | 8.3 to 20.6 inches. Outside that range is *extrapolation*: the data cannot show whether the line still holds there.

$y = \alpha + \beta x + \varepsilon$ becomes

$\text{Height} = 62.03\ (\alpha) + 1.054\ (\beta) \times \text{Girth}$

$62.03\ (\alpha) + 1.054\ (\beta) \times 15 = 77.8$ feet.

4. How sure are we?

The beta estimate, its standard error and its p-value

The 31 trees are a sample; another sample would give different estimates

4. The full report: `summary(fit)`

```
summary(fit)


#>
#> Call:
#> lm(formula = Height ~ Girth, data = trees)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -12.5816  -2.7686   0.3163   2.4728   9.9456
#>
#> Coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  62.0313     4.3833  14.152 1.49e-14 ***
#> Girth         1.0544     0.3222   3.272  0.00276 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 5.538 on 29 degrees of freedom
#> Multiple R-squared:  0.2697,  Adjusted R-squared:  0.2445
#> F-statistic: 10.71 on 1 and 29 DF,  p-value: 0.002758
```

1. **the slope: beta estimate**
   1.05 feet of height per inch
2. **standard error of the slope**
   how much the slope varies by sample
3. **t value = 1 / 2**
   3.27 standard errors away from 0
4. **p-value**
   0.0028, below 0.05: significant
5. **residual standard error**
   typical distance from the line: 5.5 feet
6. **R-squared**
   27% of the variation in height explained

The output of `summary(fit)` in the knitted R Markdown tutorial. The row `(Intercept)` gives the same four numbers for $\hat{\alpha}$.

4. The estimated equation with its standard errors and p-values

$$\text{Height} = \mathbf{62.03} + \mathbf{1.054} \times \text{Girth}$$

	$\hat{\alpha}$	$\hat{\beta}$	
standard error	(4.38)	(0.32)	how much the estimate varies from sample to sample
t value	14.15	3.27	estimate divided by its standard error
p-value	< 0.001	0.003	the chance of a value this far from 0 if the true value were 0

Decision rule

p-value below 0.05: the estimate is **statistically significant**; we conclude that the true value is not 0.

Here both p-values are below 0.05. Diameter does tell us something about height.

Interpretation

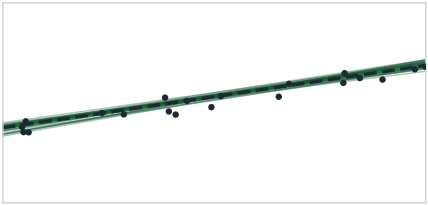
$\hat{\beta} = 1.054$, p $= 0.003$: one inch more diameter goes with 1.05 feet more height, on average, and this relationship is significant.

$\hat{\alpha}$ is significant too, but it lies outside the data and has no practical meaning here.

Camm et al., Ch. 8.5 *Testing individual regression parameters.* A significant slope can still be small: also look at its size and at R^2.

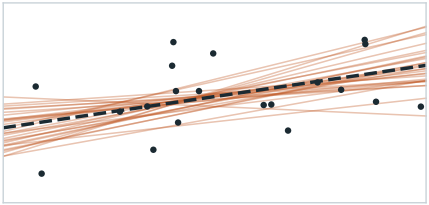
4. The standard error: what it shows

points close to the line



small standard error
(the 30 slopes have sd 0.04)

points far from the line



large standard error
(the 30 slopes have sd 0.27)

Each thin line is the fitted line of a new sample of 20 points. Dashed: the true line.

Standard error (SE) | how much the estimate would change from one sample to the next. A small standard error means a precise estimate.

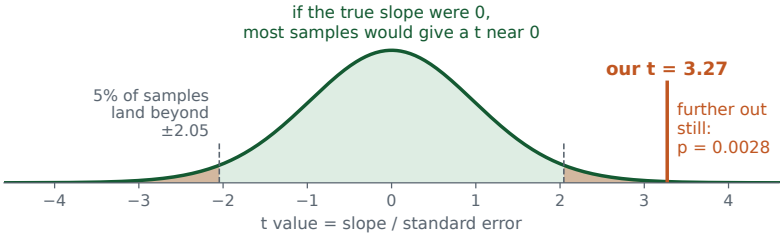
Ours | $\hat{\beta} = 1.054$ with SE $= 0.32$. Report both: "1.05 feet per inch (SE 0.32)".

The SE gets smaller with | more data, and points closer to the line.

In a plot | `geom_smooth(method = "lm")` without `se = FALSE` draws it as a grey band around the line: where the true line plausibly runs.

Camm et al., Ch. 8.5; Wright et al., Ch. 5.7.3 *Beta estimate* and *Standard error*

4. The p-value: what it shows



The question | if diameter and height had no linear relationship, the true slope would be 0. How unusual would a slope of 1.054 be then?

t value | estimate / SE = 1.054/0.32 = 3.27: the slope lies 3.3 standard errors away from 0.

p-value | the chance of getting a slope at least this far from 0 if the true slope were 0. Here 0.003: fewer than 3 samples in 1,000. So a true slope of 0 is not believable.

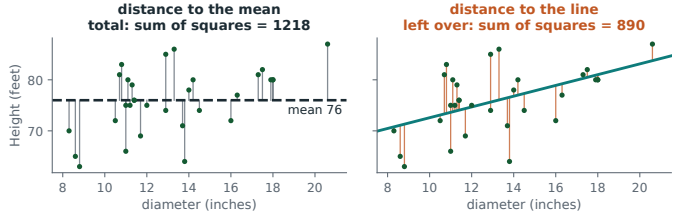
Example | a p-value of 0.30 would mean that a slope like ours turns up in 30% of samples even with no relationship: no conclusion.

The stars in `summary()`: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. The p-value is not the chance that our estimate is wrong.

5. R-squared: how much of the variation does the line explain?

Definition

R-squared is the share of the variation in y that the line explains, from 0 (nothing) to 1 (every point on the line). With one x it equals the correlation squared: $0.52^2 = 0.27$.

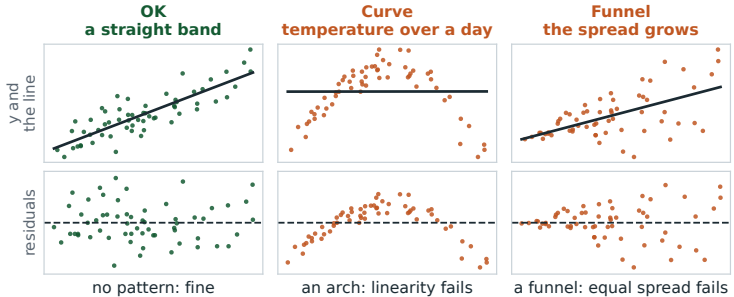


Ours: 0.27 | 27% of the differences in height between the trees go with differences in diameter. The remaining 73% is unexplained: age, soil, light and other things, the error term.

In `summary()` | `Multiple R-squared: 0.2697`. The *residual standard error*, 5.54 feet, is the typical distance of a tree from the line.

Camm et al., Ch. 8.3 *Assessing the fit*. What counts as a good R^2 depends on the field and the question.

5. The error term and the four conditions



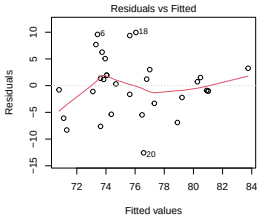
1. **Linearity**: a straight band, not a curve
2. **Independence**: one tree does not affect another
3. **Equal spread**: no funnel
4. **Normal residuals**: few large ones, roughly symmetric

Key idea

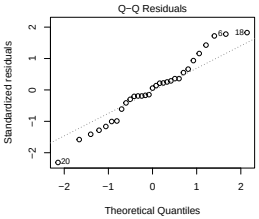
The error term collects everything that affects y apart from x. The standard error and the p-value are reliable when these four conditions hold.

Camm et al., Ch. 8.5 *Conditions necessary for valid inference*, Figures 8.16 and 8.17

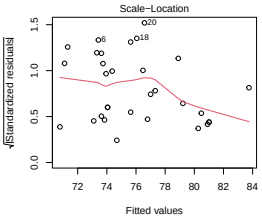
5. The four check plots



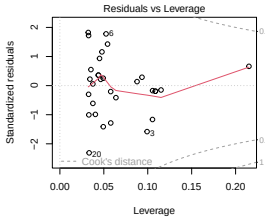
Linearity: no U or arch around the zero line



Normal residuals: points near the dotted line



Equal spread: no clear rise or fall



Influence: no point past Cook's distance lines

```
par(mfrow = c(2, 2))    # 2 by 2
plot(fit)
par(mfrow = c(1, 1))    # back to 1
```

Our trees

No U or arch; the Q-Q points follow the line; the spread falls a little on the right. With 31 trees that is mild: the conditions roughly hold. Look for clear patterns, not small wiggles.

5. The results as tables: `broom`

```
tidy(fit)


#> # A tibble: 2 x 5
#>   term        estimate std.error statistic    p.value
#>   <chr>          <dbl>     <dbl>     <dbl>      <dbl>
#> 1 (Intercept)  62.031    4.3833    14.152  1.4930e-14
#> 2 Girth         1.0544   0.32222    3.2722 2.7578e- 3

glance(fit)


#> # A tibble: 1 x 12
#>   r.squared adj.r.squared  sigma statistic   p.value    df  logLik    AIC    BIC
#>       <dbl>         <dbl>  <dbl>     <dbl>     <dbl> <dbl>   <dbl>  <dbl>  <dbl>
#> 1   0.26965       0.24447 5.5385    10.707 0.0027578     1 -96.017 198.03 202.34
#> # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

The output in the knitted R Markdown tutorial

`tidy(fit)` | the estimates as a table: estimate, standard error, t value (`statistic`) and p-value. The slope's row is boxed.

`glance(fit)` | one row about the whole model: R-squared (boxed), `sigma` (the residual standard error), p-value.

`augment(fit)` | one row per tree, with `.fitted` and `.resid`.

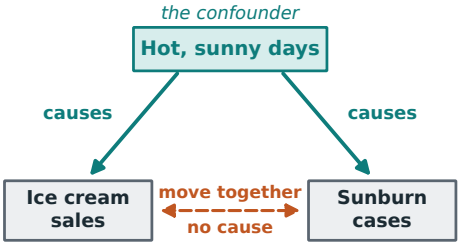
Why use them | the same numbers as `summary()`, but as tibbles that work with the dplyr verbs.

6. Correlation and causation

What a regression can and cannot say

Association does not imply causation

6. Correlation is not causation



Key idea

A regression measures how x and y move together. On its own it says **nothing about cause and effect**.

Camm et al., Ch. 8.5 (p. 412); Wright et al., Ch. 5.7.11 and 5.7.12

Confounder | a third variable that drives both x and y. Thicker trees tend to be taller, but a thicker trunk does not make a tree grow taller: older trees are both thicker and taller, so *age* drives both.

Reverse causation | y may drive x.

Coincidence | among many variables, some pairs correlate by chance: margarine consumption in the US and the divorce rate in Maine, $r = 0.99$.

To show a cause | a controlled experiment, where only x is changed on purpose.

Check your understanding

1. In `lm(sales ~ advertising, data = shop)`, which variable is the dependent variable?

2. A slope is 2.5 with a standard error of 0.5. What is the t value? Is the slope far from 0?

3. A slope has a p-value of 0.30. What do you conclude?

4. $R^2 = 0.64$ and the slope is positive. What is r?

5. The more fire fighters come to a fire, the more damage it does. Do the fire fighters cause the damage?

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4. $R^2 = 0.64$ and the slope is positive. What is r?

5. The more fire fighters come to a fire, the more damage it does. Do the fire fighters cause the damage?

Answers

1. `sales`: the variable left of the tilde.

2. $t = 2.5/0.5 = 5$. Five standard errors from 0: the p-value is tiny.

3. No evidence of a linear relationship: a true slope of 0 is plausible. That is not proof that there is none.

4. $r = +0.8$, the square root of 0.64 with the sign of the slope.

5. No. Big fires bring many fire fighters *and* cause much damage: the size of the fire is the confounder.

What we covered today

Concepts

- ✓ predictive and descriptive data mining
- ✓ the model $y = \alpha + \beta x + \varepsilon$, least squares, residuals
- ✓ the estimates $\hat{\alpha}$ and $\hat{\beta}$ in the units of the data
- ✓ standard error, t value, p-value, the decision rule
- ✓ R-squared, the error term and the four conditions
- ✓ correlation is not causation; confounders

Tools

- ✓ `data(trees)`, `lm()`, the formula `y ~ x`
- ✓ `geom_smooth(method = "lm")`
- ✓ `fitted()`, `residuals()`, `predict()`
- ✓ `summary()`
- ✓ `plot(fit)` with `par(mfrow = c(2, 2))`
- ✓ `tidy()`, `glance()`, `augment()` from `broom`

Objective: fit a simple linear regression, interpret the estimates, and judge their precision and the fit.

Before the next lecture

To do

✓ finish the script if you did not

✓ read Camm et al., Ch. 8, sections 8.1 to 8.3 and 8.5, and Wright et al., Ch. 5.7

Next lecture

Tuesday 3 November.

The thread

Lecture 7: describe data with numbers.
Lectures 8 and 9: visualise data.
Lecture 10: model one variable with another.

Sources used in this lecture

Course reading

- Camm, J. D., Cochran, J. J., Fry, M. J. & Ohlmann, J. W. (2024). *Business Analytics* (5th ed.). Cengage. Ch. 1.3 and the introductions to Ch. 6 and Ch. 10: data mining. Ch. 8 *Linear Regression*, sections 8.1 to 8.3 and 8.5.
- Wright, C., Ellis, S., Hicks, S. & Peng, R. (2021). *Tidyverse Skills for Data Science in R*, Ch. 5.7 *Linear Modeling*. https://jhudatascience.org/tidyversecourse/model.html#linear-modeling

Data and packages

- `trees`: Ryan, T. A., Joiner, B. L. & Ryan, B. F. (1976). *The Minitab Student Handbook*. Duxbury. In R: `?trees`.
- broom: Robinson, D., Hayes, A. & Couch, S. Convert statistical objects into tidy tibbles. https://broom.tidymodels.org/

Course material

- GitHub: https://github.com/touseefhameed/oka2015-applied-data-science Google Colab: Open in Colab

Thank you



Course materials: `github.com/touseefhameed/oka2015-applied-data-science`

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